

Assignment 2

Designing a Study: From Question to Identification

What this assignment is. This is a **design** assignment — you will not estimate a treatment effect here. Instead you build the scaffolding for a credible study around a **research question you care about** (ideally the one for your course paper). You will motivate it, hunt for data, specify the study as a **target trial** you would run if you could, translate that into an **applied-HSR identification strategy**, and finally use AI to plan — and debug — the data cleaning it would require. **Submit** a knitted R Markdown file. Most of it is prose and one figure (a DAG); the only code is in Section 5. Include the AI-use statement. Late penalty 10%/day.

Video (★). The ★ part in Section 3 requires a video of no more than 3 minutes explaining your identification strategy and its single biggest threat. See the syllabus for details.

A note on scope. *Pick a question narrow enough that you can actually name the data and the variation that would identify it. “Does health insurance affect health” is too big; “did the 2014 Medicaid expansion change ED visits among low-income adults in expansion vs. non-expansion states” is the right grain.*

Section 1 — Motivation & Research Question

This is your framing. Write it yourself. This doesn’t have to be more than 5 paragraphs total, I am not expecting a full paper here.

1. **State the causal question precisely.** Name the treatment, the outcome, and the population: “What is the effect of ___ on ___ among ___?” Be as clear as possible when discussing your estimand.
2. **Motivate it from (at least one of) three vantage points.** A clinician's, a researcher's, and a policy-maker's. What decision would these groups make differently once they knew the answer?
3. **Why isn't this already settled?** What makes the question hard — is it confounding, selection, measurement, or simply that no one has the right data?
4. **What data would be useful?** Good empirical work is often bottlenecked by data, not ideas. Identify at least two candidate data sources for your project. For each, specify: the unit of observation, coverage and time span, how you would obtain the treatment, outcome, and key confounders, access/feasibility (public, licensed, RTA), and its main limitation. Which one would you prefer and why?

Section 2 — Target Trial Emulation

Before choosing a research design, specify the **randomized experiment you would run if you could** — the “target trial” — and then describe how you would emulate it with the observational data from Section 1. This discipline forces the estimand to be explicit and surfaces alignment problems (especially around *time zero*) before they contaminate your estimates.

2.1 Specify the target trial and its emulation

Complete the protocol below for your study. In each row, describe both the ideal-trial version and how you emulate it in your data (you can note the emulation in the same cell).

Target-trial component	What to specify for your study (fill in)
Eligibility criteria	Who would enroll? Translate into concrete inclusion/exclusion rules you can apply in data.
Treatment strategies	The interventions being compared (e.g., “start drug X” vs. “no drug X”). Be precise about what “treated” means.
Assignment procedure	In the ideal trial, randomized. In your emulation, how is treatment assigned, and what makes it non-random?
Time zero (start of follow-up)	When does the clock start for each unit? How do you align it for treated and comparison groups? Flag any immortal-time risk.
Outcome	The measured outcome and its timing window.
Causal contrast	The estimand — intention-to-treat vs. per-protocol, ATE vs. ATT vs. LATE.
Analysis plan	The comparison you would run (kept high-level here — you formalize the method in Section 4).

2.2 Time zero and immortal time. In 3–5 sentences, state exactly when follow-up begins for treated and comparison units, and explain how a careless alignment (e.g., classifying someone as “treated” using information that only arrives *after* baseline) would create **immortal time bias**. You will return to this in Section 4.

Section 3 — The Applied Research Design

Now translate the target trial into the language of applied causal inference: identify the **source of variation** that stands in for randomization, and the method that exploits it.

1. **Name your identification strategy.** What is the variation that identifies the effect (a policy change, an eligibility threshold, an instrument, a natural experiment, selection-on-observables)? Which method follows (DiD, event study, IV, matching, ...), and why is it the right fit for *this* target trial? [Note: we haven’t covered these methods in detail yet, so I am not expecting you to do so at this stage either.]
2. **Draw the DAG.** Diagram the treatment, outcome, and the confounders/mediators/colliders that matter. Mark which are observed and which are not (hand-drawn and scanned is fine).
3. **State the identifying assumption in one sentence.** Complete precisely: “This design recovers the causal effect only if ...,” then give one reason to believe it and one reason to doubt it in your setting.
4. **Name the threats and one test.** What are the leading threats (confounding, selection, reverse causality, spillovers)? Propose one falsification or placebo test you could run to probe your main assumption.

5. ★ **Reflect: what does each framing add?** In your video, explain your identification strategy and its single biggest threat. Comment on what the *target-trial* framing (Section 2) made visible that the *causal inference* framing did not, or vice versa.

Section 4 — AI-Assisted Data Cleaning & Debugging

4.1 Plan the cleaning with AI (use it)

Using the dataset you chose in Section 2, use an AI assistant to produce a **concrete, ordered list of data-cleaning steps** that would turn the raw data into the analytic sample your design needs. These might include (although this list is not exhaustive): operationalizing eligibility, constructing the treatment and *time zero*, building the outcome, handling missing data, and applying sample restrictions.

- **Submit the prompt** you gave the AI, its **raw list**, and then **your edits**: what did it miss or get wrong? Did it add something you had failed to consider that you think is right?
- You do not need to run this on real data.

4.2 Debug the AI's cleaning script (catch what it gets wrong)

A student asked an AI assistant to write the cleaning script below for a generic longitudinal dataset (patient visits, a treatment file, and demographics). **It runs without any error message**, but it has **at least three consequential problems** — some are coding bugs, some would bias *any* downstream design. Find and fix them: for each, state (i) what is wrong, (ii) the corrected code, and (iii) how it would distort the study.

```
library(tidyverse)
visits  <- read_csv("visits.csv")      # one row per patient-visit
patients <- read_csv("patients.csv")   # one row per patient (demographics)
tx      <- read_csv("treatment.csv")   # treatment start dates (treated only)

# 1. Parse dates
visits <- visits %>% mutate(visit_date = as.Date(visit_date, format = "%m/%d/%Y"))
tx     <- tx      %>% mutate(tx_date   = as.Date(tx_date))   # format unspecified

# 2. Attach treatment info to visits
df <- visits %>% inner_join(tx, by = "patient_id")

# 3. Define treatment and start of follow-up
df <- df %>% mutate(treated = 1,
                   time_zero = min(visit_date))

# 4. Restrict to post-baseline visits with a complete outcome
df <- df %>% filter(!is.na(outcome)) %>%
  filter(visit_date >= time_zero)

# 5. Add demographics
df <- df %>% left_join(patients, by = "patient_id")
```

Section 5 — Mini Referee Report

Peer review is the part of the job no AI can do for you. This capstone turns the design lens you just built on someone else's work.

Assigned paper: You may choose one paper from three options posted in our GitHub folder under **HAD5744\Lecture06_Matching\RefereeReportOptions**.

Read the assigned paper and write a short referee report (about one page, 400–600 words) naming the **two or three MAJOR concerns** — the ones that would change your confidence in the main result — not minor issues of writing. Structure it as: (1) a 2–3 sentence summary of the claim and its identification strategy; (2) two or three major concerns, each with why it threatens the central result and what would resolve it; and (3) a verdict naming the single most important revision.

Focus for this assignment: read the paper as a design critic, using the tools from Sections 3 and 4. Is the causal question well-posed? Can you reconstruct the target trial it emulates — and is time zero aligned for the treated and comparison groups, or is there immortal-time or look-ahead risk? Is the source of identifying variation credible, is the estimand stated, and does it match the claim the paper actually makes?

Using AI here. You may use AI to understand the paper, but the concerns must be your own. Its first list of concerns is usually generic and agreeable; referee value comes from the concern specific to this paper and its design. Note in one line what it suggested and whether you agreed.